

The dot plot, histogram, and box plot summarize the hours of battery life for 26 cell phones that are constantly streaming videos.

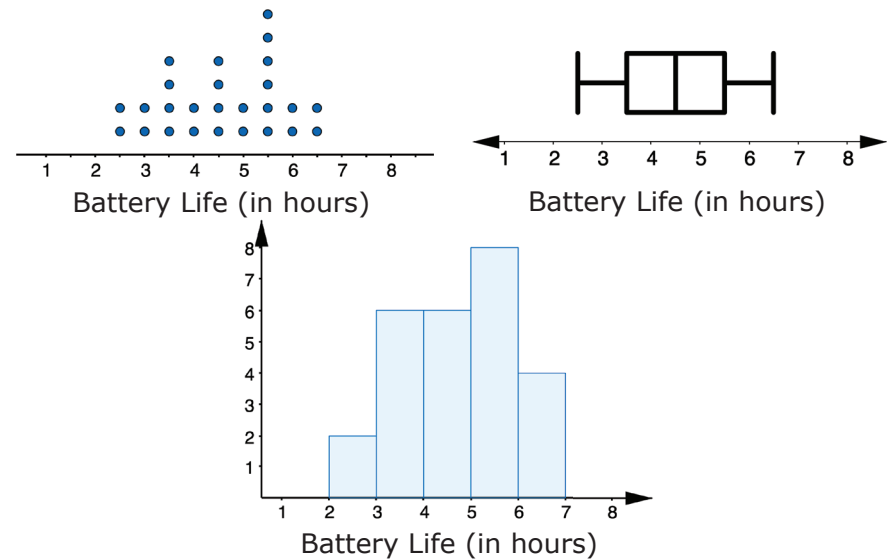


MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.



- ☐ I can justify whether or not a measure of center or variation appropriately summarizes data.



1. What do you notice?
2. What do you wonder?



10.3.2 Collaboration: Comparing Data Displays

Suppose an office tracks the number of phone calls received on business days, Monday through Friday, over two weeks. The data set is shown below.

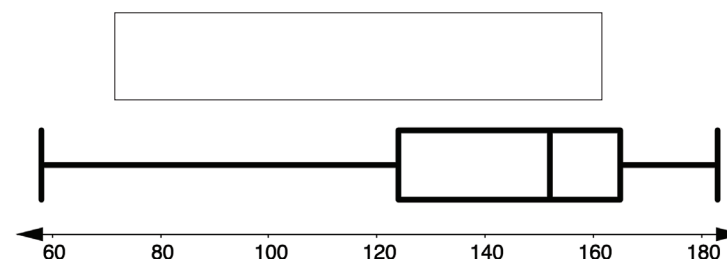
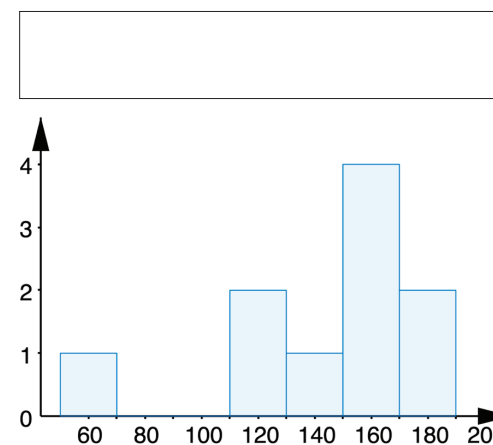
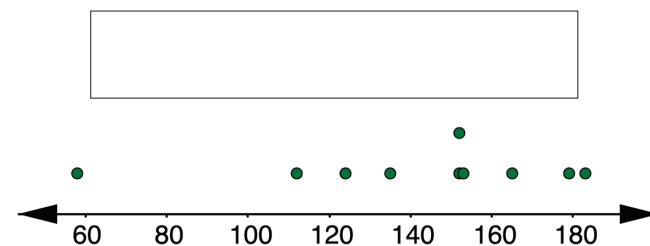
183	165	58	152	179
152	153	135	124	112

- Discuss with your partner what you notice about the data set.
- Match the measures of center and variation with their calculations for the data set above.

A. Mean	B. Median	C. Range	D. IQR
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	$\frac{152+152}{2} = 152$
	$165 - 124 = 41$
	$\frac{183 + 165 + 58 + 152 + 179 + 152 + 153 + 135 + 124 + 112}{10} = 141.3$
	$183 - 58 = 125$

- A **histogram**, **line plot**, and **box plot** representing the data are shown.



- Label the display types by filling in the box above each display.

- b. Complete the table by identifying which measures of center and variation can be determined from each display.

	Mean	Median	Range	IQR
Histogram	☐	☐	☐	☐
Line Plot	☐	☐	☐	☐
Box Plot	☐	☐	☐	☐

4. Describe any outliers in the data set.

5. Explain which of the displays you used to determine whether or not there was an outlier in the data set.



10.3.3 Guided Instruction: Stem-and-Leaf Plots

Recall the data collected on the number of phone calls an office received on business days over two weeks.

183	165	58	152	179
152	153	135	124	112

In addition to displaying the data graphically with a histogram, line plot, or box plot, the data can also be displayed in a **stem-and-leaf** plot.



A **stem-and-leaf plot** is a table that organizes data by place value to compare data frequencies.

1. What do you notice about the data in the stem-and-leaf plot?

Stem	Leaf
5	8
6	
7	
8	
9	
10	
11	2
12	4
13	5
14	
15	2, 2, 3
16	5
17	9
18	3

Key: 5|8 = 58

2. Compare the stem-and-leaf plot with the line plot from the collaboration.

What similarities do you notice between the displays?	What differences do you notice between the displays?

3. What value does the entry 13|5 represent?

4. Which measure(s) of center can be determined from the stem-and-leaf plot?

- ☐ mean
- ☐ median
- ☐ both the mean and median

5. Which measure(s) of variation can be determined from the stem-and-leaf plot?

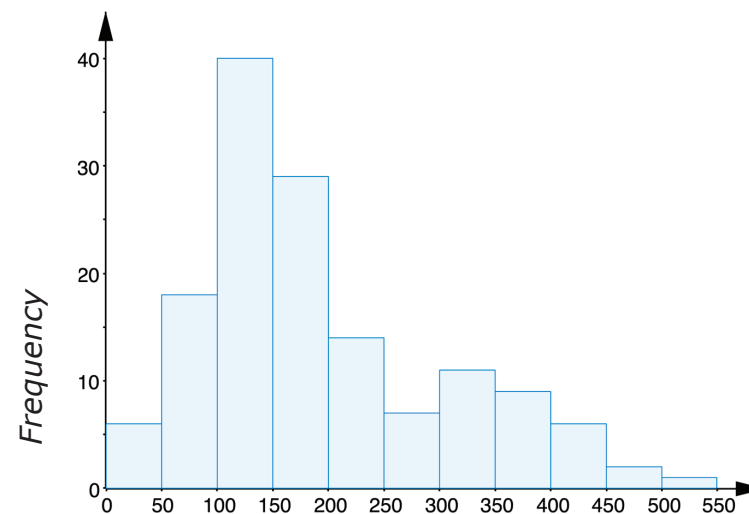
- ☐ range
- ☐ interquartile range
- ☐ both the range and the interquartile range



10.3.4 Collaboration: Center and Variation in Context

1. In a study of wild bears, researchers measured the weights, in pounds (lb), of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.

Wild Bear Weight



Weight (pounds)

- a. Complete the statements.

Based on the data, it appears wild bears typically

weigh between ☐ 50 and 150 ☐ 100 and 200 ☐ 150 and 250 pounds. A good

estimate for the typical weight for wild bears is

_____ pounds.

- b. Compare your estimate with your partner. Discuss whether it is possible for both of your estimates to be reasonable, even if they are not the same.

- c. Discuss with your partner whether your estimate from Part A is closer to the median or the mean. Summarize your discussion.

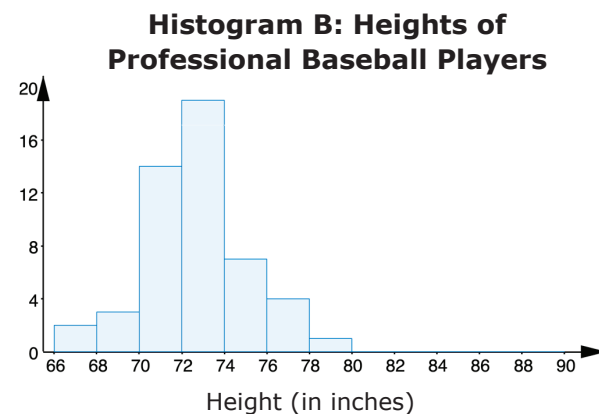
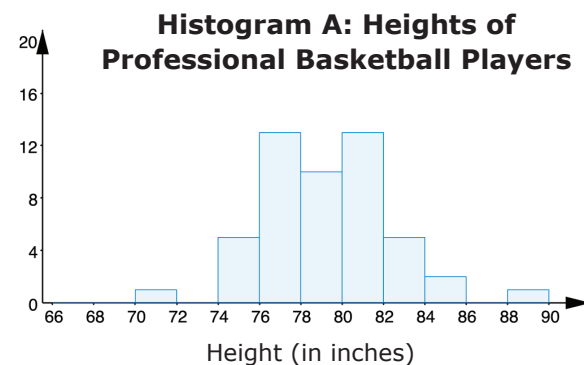
- d. Angelica and Raymond were asked to describe the variation in weight of the wild bears.

Angelica said, "The bear weights range from less than 50 to over 500 lbs, so the range best describes the spread of the weights of the wild bears."

Raymond said, "It looks like most bears in the sample weigh between 100 and 300 lbs, with few weighing less and few weighing more. The small weights could be because there were babies in the sample. So, the interquartile range is the best measure of spread to describe this data."

Explain whether you agree with Angelica or Raymond. Justify your reasoning.

2. The two histograms, labeled A and B, show height distributions of 50 male professional basketball players and 50 male professional baseball players.



Work with your partner to complete each statement.

- a. In ☐ Histogram A,
☐ Histogram B, the average height is around 72 inches. The values of the mean and median are similar.
- b. To explain typical height, the variation in ☐ Histogram A
☐ Histogram B is best described by the interquartile range because the range may be misleading due to outliers. The values of the mean and median are similar.

3. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer displayed the earthworms' lengths, in millimeters (mm), in a stem-and-leaf plot.

Stem	Leaf
0	6
1	1, 8, 9
2	0, 3, 3, 5, 5, 6, 7, 7, 8, 9
3	2, 3
4	1, 2, 8
5	2, 4, 9
6	0
7	7
8	
9	3

Use the display to answer the following questions.

Key: $2|8 = 28$

- Half of the earthworms sampled were less than _____ mm long.
- The farmer reported that the average length is 35.92 mm. Why might this measure of center be misleading about the typical length of the earthworms in this sample?

- Compare your answers to Parts A and B with your partner and, if necessary, resolve any differences.
- The farmer reported that the lengths varied between 6 mm and 93 mm. The farmer's assistant said that most earthworms had lengths between 23 and 48 mm.

Work with your partner to complete the statements.

The farmer is using the ☐ IQR ☐ range to summarize

the variation in worm length, while the farmer's

assistant is using the ☐ IQR. ☐ range.

If a researcher is interested in how long

earthworms are, the ☐ IQR ☐ range is an

appropriate measure of variation for the farmer

to report because _____

_____.



10.3.5 Wrap-Up: Reflection

Draw an **X** on the line to indicate where your current learning is in relation to the following learning targets.

1. I can justify whether or not a measure of center appropriately summarizes data.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.



2. I can justify whether or not a measure of variation appropriately summarizes data.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.



Unit 10, Lesson 4: Determining Appropriate Measures of Center and Variation



Benchmark

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, while taking into consideration the context and any outliers.



Learning Target

- ☐ I can determine an appropriate measure of center and variation to summarize numerical data.